



Required Capstone Project: Module 11

Due Aug 19 by 11:59pm**Points** None**Available** after Aug 12 at 12am

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**Learning Outcomes Addressed**

- Understand how AI, GenAI and Agentic AI work and where they apply in business.
- Use prompts effectively for analysis, decision-making and content generation.
- Map business workflows and identify where AI can drive automation and efficiency.
- Evaluate AI opportunities using feasibility, ROI and impact criteria.
- Build AI adoption roadmaps aligned to business goals and KPIs.
- Apply data governance, privacy and ethical AI principles.

Phase 5 – Deployment & Governance

Milestone 11: Enterprise Architecture & Vendor Strategy

Estimated Effort: 2 Hours

Objective

Designing an AI solution requires more than conceptual ideas.

Organisations must also determine **how the solution will be deployed and operated in practice**.

This includes decisions about:

- which technology platforms and vendors to use
- how the AI system integrates with existing enterprise infrastructure
- how data flows through the architecture
- how reliability, security, and scalability are maintained

Most organisations implement AI solutions using **ecosystems of platforms, APIs, data systems, and cloud infrastructure**, rather than building everything from scratch.

In this module, you will design a **high-level enterprise architecture and vendor strategy** for deploying the AI solution developed in your capstone project.

How This Connects to Earlier Modules

In earlier modules, you designed the **functional logic of your AI system**, including:

- workflow redesign
- prompt architecture
- agent architecture

- automation logic and governance
- strategic and financial validation

In this module, you move from **conceptual design to deployment thinking**.

You will define:

How would this AI system actually be implemented in a real organisation?

Note: This is a required assignment and counts towards programme completion.

Task Overview

Create an **Enterprise Architecture & Vendor Strategy** for your AI solution.

Your design should define the following components:

1. Vendor ecosystem options
2. Build vs buy vs partner decision
3. Data architecture
4. Integration model
5. Security layer
6. Budget range estimate

The goal is not to design a detailed technical architecture, but rather to define a **clear and realistic deployment strategy**.

Step 1 — Define the Vendor Ecosystem

Most AI solutions rely on **multiple vendors and platforms working together**.

Your vendor ecosystem should identify the types of platforms required to implement your solution.

Typical components may include:

- cloud infrastructure providers
- AI model providers

- data storage platforms
- orchestration frameworks
- enterprise software integrations

Example Vendor Ecosystem (Illustrative)

Layer	Example Role
Cloud Platform	Provides computing infrastructure
Foundation Model Provider	Provides generative AI model APIs
Workflow Automation Platform	Manages AI task orchestration
Data Platform	Stores and processes organisational data

Your response should identify **the types of platforms needed**, rather than specific vendor contracts.

Step 2 — Decide Build vs Buy vs Partner

Organisations must decide whether to:

Build

Develop the AI solution internally using internal engineering teams.

Buy

Adopt an existing commercial AI product or software platform.

Partner

Collaborate with technology vendors or consulting firms to implement the solution.

Most enterprise AI systems combine **multiple approaches**.

Example Decision (Illustrative)

Component	Approach
AI Model Access	Buy API access from model provider
Workflow Automation	Build internal orchestration layer
Integration Services	Partner with technology vendor

Your explanation should briefly justify **why each approach makes sense for your solution.**

Step 3 — Define the Data Architecture

AI systems rely heavily on **organisational data**.

Your data architecture should explain:

- where the data originates
- how it is stored
- how the AI system accesses it
- how data flows through the workflow

Consider data sources such as:

- enterprise databases
 - CRM systems
 - ERP platforms
 - internal documents or knowledge bases
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Example (Illustrative)

Customer enquiries originate from the customer support platform, are processed by the AI agent, and relevant historical customer data is retrieved from the CRM system.

Step 4 — Define the Integration Model

AI systems must integrate with existing **enterprise applications and workflows**.

Your integration model should explain:

- which enterprise systems the AI solution connects with
 - how data is exchanged between systems
 - whether integration occurs through APIs, middleware, or automation tools
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Example

System	Integration Purpose
CRM Platform	Retrieve customer data

System	Integration Purpose
Support Platform	Receive incoming support tickets
Analytics Platform	Track AI performance metrics

This ensures that AI systems operate **within existing organisational processes**.

Step 5 — Define the Security Layer

AI systems must operate within organisational **security and governance frameworks**.

Your architecture should consider basic security components such as:

- authentication and access control
- data privacy protection
- monitoring and audit logs
- secure API communication

Security ensures that the AI system operates **safely and responsibly within enterprise environments**.

Step 6 — Estimate the Budget Range

Finally, estimate a **high-level budget range** required to implement the AI solution.

This does not require exact pricing, but should include approximate categories such as:

- AI model API costs
 - cloud infrastructure costs
 - integration and development effort
 - software licensing or vendor subscriptions
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Example Budget Estimate (Illustrative)

Cost Category	Estimated Annual Cost
AI Model API Usage	£20,000 – £50,000
Cloud Infrastructure	£10,000 – £30,000
Integration Development	£40,000 – £80,000

Cost Category	Estimated Annual Cost
Monitoring and Security Tools	£5,000 – £15,000

Your estimate should reflect the **scale of the AI solution you designed earlier in the capstone.**

Required Output Structure

Your **Enterprise Architecture & Vendor Strategy** should include:

1. AI Architecture Overview

Brief explanation of the overall system architecture.

2. Vendor Ecosystem

Description of the platforms or types of vendors required.

3. Build vs Buy Decision

Explanation of which components will be built internally or sourced externally.

4. Data and Integration Architecture

Explanation of data sources, data flows, and system integrations.

5. Security Layer

Overview of how security and access control are handled.

6. Estimated Implementation Budget

High-level estimate of implementation and operational costs.

Practical Tips

Keep the Architecture High-Level

You are not expected to design a full technical architecture diagram.

Focus on **major architectural components and decisions**.

Align Architecture With Your Workflow Design

Your architecture should clearly support the **AI workflow and agents designed in earlier modules**.

Consider Realistic Enterprise Constraints

Think about:

- existing enterprise systems
- data availability
- operational security requirements

Enterprise AI solutions must integrate with **real organisational environments**.

Avoid These Common Mistakes

- ✗ Designing an architecture that ignores existing enterprise systems
- ✗ Assuming all components must be built internally
- ✗ Ignoring security or governance considerations
- ✗ Providing unrealistic cost estimates

Effective AI deployment strategies balance **technical capability, operational integration, and cost efficiency**.

Submission

Before submitting your **Enterprise Architecture & Vendor Strategy**, confirm that you have included:

- ✓ High-level AI architecture overview
- ✓ Vendor ecosystem description
- ✓ Build vs buy vs partner decision
- ✓ Data and integration architecture
- ✓ Security considerations
- ✓ Budget range estimate

Instructions:

- Use the same document that you downloaded in Module 1 and include your responses.
- We recommend you add additional text boxes or slides (if needed) for each milestone task to ensure that your responses are documented in a proper format.
- The document will gradually grow as you complete each capstone milestone. The final submission occurs at the end of Module 14.

Grading Rubric

Maximum Points: 20

Criteria	Full Points	0 Points
AI Architecture Overview (3 points)	The submission explains the overall AI architecture and describes how the proposed solution will be deployed within an enterprise environment. (3 points)	The AI architecture overview is missing. (0 points)
Vendor Ecosystem (3 points)	The submission identifies the required vendor ecosystem and explains the role of the different platforms or technology providers needed to support the AI solution. (3 points)	The vendor ecosystem and the role of the required platforms is not identified. (0 points)

Criteria	Full Points	0 Points
Build vs Buy vs Partner Decision (4 points)	The submission explains the Build vs Buy vs Partner decisions for the AI solution and justifies the selected approach for the major components. (4 points)	The Build vs Buy vs Partner decisions or supporting justification are missing. (0 points)
Data & Integration Architecture (4 points)	The submission explains the data architecture, data flow, and integration with enterprise systems required to support the AI solution. (4 points)	The data architecture and system integration is missing. (0 points)
Security Layer (3 points)	The submission explains the security considerations, including access control, data protection, and governance mechanisms required for the AI solution. (3 points)	Security considerations and governance mechanisms are missing. (0 points)
Estimated Implementation Budget (3 points)	The submission provides a high-level implementation budget, including the major cost categories required to deploy the AI solution. (3 points)	The implementation budget and major cost categories are missing. (0 points)

Why this Step Matters

Many AI initiatives fail not because the technology is ineffective, but because organisations struggle to **deploy and integrate AI solutions into existing systems**.

Successful AI deployment requires:

- clear architectural design
- appropriate vendor partnerships
- strong data infrastructure
- reliable integration with enterprise workflows

Understanding how to design **deployable AI architectures** is a critical capability for organisations implementing AI at scale.

In the next module, you will design an **AI governance and risk management framework** to ensure your solution can operate responsibly within organisational and regulatory environments.